



SACE STAGE 2 · REVISION CHECKLIST

Physics

Every topic you need to know, in one checklist.

3 topics

50 checkpoints

TopMark Tutors

How to use this checklist

Work through every line and tick honestly. Do the whole thing in one sitting before you revise anything — that first pass becomes your study plan for the term.

1

Revised

You have gone back over the content and understand it.

2

Practised

You have done real exam questions on it, not just read notes.

3

Confident

You could do it in the exam, under time, with no notes.

Most students stop at column 1. Re-reading notes feels like studying, but marks are won in columns 2 and 3. Anything ticked in column 1 and blank in column 3 is exactly where your marks are leaking.

Externally assessed by written examination (30% of final grade).

Produced by TopMark Tutors as a free study resource, structured around the SACE Stage 2 Physics subject outline. It is a revision aid only and is not produced, endorsed, or reviewed by the SACE Board of South Australia. Always check the current subject outline for your enrolment year and confirm content with your teacher.



Topic 1 — Motion and Relativity

TOPIC 1 OF 3

1.1 Projectile Motion

CONTENT	REVISED	PRACTISED	CONFIDENT
Resolving initial velocity into horizontal and vertical components	☐	☐	☐
Independence of horizontal and vertical motion	☐	☐	☐
Calculating time of flight, range and maximum height	☐	☐	☐
Projectiles launched from a height, and non-symmetrical trajectories	☐	☐	☐
Sign conventions and why this 'easy' topic loses so many marks	☐	☐	☐

1.2 Forces and Momentum

Newton's laws applied to systems of connected bodies	☐	☐	☐
Impulse, change in momentum, and force–time graphs	☐	☐	☐
Conservation of momentum in elastic and inelastic collisions	☐	☐	☐
Collisions in two dimensions	☐	☐	☐

1.3 Circular Motion and Gravitation

Centripetal force and acceleration, and identifying which force supplies it	☐	☐	☐
Vertical and horizontal circular motion, including banked tracks	☐	☐	☐
Newton's law of universal gravitation	☐	☐	☐
Satellite motion, orbital period and geostationary orbits	☐	☐	☐



Topic 1 — Motion and Relativity (continued)

TOPIC 1 OF 3

1.4 Relativity

Inertial reference frames and the postulates of special relativity

☐☐☐

Time dilation and length contraction calculations

☐☐☐

Evidence for relativity (e.g. muon decay)

☐☐☐Mass–energy equivalence, $E = mc^2$ ☐☐☐**WEAK SPOTS TO COME BACK TO**

**Topic 2 — Electricity and Magnetism****TOPIC 2 OF 3****2.1 Electric Fields**

CONTENT	REVISED	PRACTISED	CONFIDENT
Coulomb's law and field strength around point charges	☐	☐	☐
Field patterns for point charges and parallel plates	☐	☐	☐
Relationship between field strength, potential difference and plate separation	☐	☐	☐

2.2 Motion of Charged Particles in Electric Fields

Work done and energy change on a charged particle	☐	☐	☐
Acceleration of charges and projectile-like motion between plates	☐	☐	☐
Electron-volt as a unit of energy	☐	☐	☐

2.3 Magnetic Fields

Field patterns around wires, loops and solenoids	☐	☐	☐
Force on a current-carrying conductor, $F = BIL \sin\theta$	☐	☐	☐
Using the right-hand rule to determine direction	☐	☐	☐

2.4 Motion of Charged Particles in Magnetic Fields

Force on a moving charge, $F = qvB \sin\theta$	☐	☐	☐
Circular paths of charges in magnetic fields and radius calculations	☐	☐	☐
Applications: the cyclotron, mass spectrometer, velocity selector	☐	☐	☐



Topic 2 — Electricity and Magnetism (continued)

TOPIC 2 OF 3

2.5 Electromagnetic Induction

Magnetic flux and Faraday's law of induction	☐	☐	☐
Lenz's law and determining the direction of induced current	☐	☐	☐
EMF induced in a moving conductor and in a rotating coil	☐	☐	☐
Transformers and their operating principle	☐	☐	☐

WEAK SPOTS TO COME BACK TO



Topic 3 — Light and Atoms

TOPIC 3 OF 3

3.1 Wave Behaviour of Light

CONTENT	REVISED	PRACTISED	CONFIDENT
Young's double-slit experiment and calculating fringe spacing	☐	☐	☐
Diffraction gratings and the grating equation	☐	☐	☐
Interference conditions for constructive and destructive interference	☐	☐	☐
Evidence that light behaves as a wave	☐	☐	☐

3.2 Wave-Particle Duality

The photoelectric effect and why it cannot be explained by wave theory	☐	☐	☐
Work function, threshold frequency and stopping voltage	☐	☐	☐
Interpreting a photoelectric graph to find the work function and Planck's constant	☐	☐	☐
Photon energy calculations, $E = hf$	☐	☐	☐
De Broglie wavelength and evidence for matter waves	☐	☐	☐

3.3 Structure of the Atom

Emission and absorption spectra and their formation	☐	☐	☐
Energy levels, electron transitions and calculating photon energy	☐	☐	☐
Why each element has a unique spectrum	☐	☐	☐
Evidence supporting atomic models	☐	☐	☐



Topic 3 — Light and Atoms (continued)

TOPIC 3 OF 3

3.4 The Standard Model

Quarks, leptons and the composition of hadrons (baryons and mesons)

☐☐☐

The four fundamental forces and their exchange particles

☐☐☐

Conservation laws: charge, baryon number, lepton number

☐☐☐

Determining whether a proposed particle interaction is permitted

☐☐☐**WEAK SPOTS TO COME BACK TO**



WHAT TO DO WITH THE BLANKS

The gaps you just found are the whole point.

Every line you could not tick in the 'Confident' column is a mark you are currently at risk of dropping. Our SACE Exam Crash Course is built to close exactly those gaps — we mapped four years of real SACE papers to find where marks are actually lost, then weighted the teaching toward those points instead of re-covering content you already know.

The SACE Exam Crash Course — Physics

- 28 September – 10 October · Methods, Chemistry, Physics, Biology
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- Finishes with a full mock exam, marked and returned with your personal mark-loss breakdown
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